

KALYANI M G

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RESEARCH PROFILE

Third-year B.Tech Computer Science & Engineering student (CGPA: 9.27/10) at LBS Institute of Technology for Women, with hands-on research experience in machine learning, NLP, and recommendation systems. Demonstrated ability to design, implement, and evaluate ML models across real-world problem domains. Passionate about advancing AI research through rigorous experimentation, technical writing, and embedding-based modelling.

RESEARCH EXPERIENCE

Research Intern | College of Engineering Thiruvananthapuram (CET)

June 2025 – June 2025

- Participated in the "Innovative Design and Experimentation of Algorithms (IDEA)" internship under faculty mentorship.
- Designed and developed a content & genre-based **cross-domain recommendation system** for books and movies.
- Applied TF-IDF similarity, cosine similarity, and reviewer overlap analysis; implemented multiple filtering strategies to balance accuracy and diversity.
- Conducted structured performance experiments, literature review, and produced formal technical documentation toward academic publication.

Machine Learning Intern | CodSoft

Dec 2025 – Jan 2026

- Built and evaluated 3 end-to-end ML pipelines: movie genre classifier (DistilBERT/NLP), customer churn predictor (Gradient Boosting), and SMS spam detector (SVM + TF-IDF).
- Strengthened skills in data preprocessing, feature engineering, model selection, hyperparameter tuning, and evaluation metrics (F1, AUC-ROC).
- Deployed interactive inference interfaces using Streamlit and Hugging Face Transformers.

PROJECTS

Cervical Cancer Detection System — Python, Deep Learning, CNN, TensorFlow/PyTorch

- Developed a deep learning-based medical image classification system for cervical cancer detection using cervical cell images.
- Implemented image preprocessing, augmentation, feature extraction, and CNN-based classification to identify cancer-related patterns.
- Built an interactive prediction interface to support real-time image analysis and early disease detection.

Cross-Domain Recommendation System — Python, TF-IDF, Cosine Similarity, Pandas, Scikit-Learn

- Built a recommendation engine spanning books and movies by combining genre matching, TF-IDF text similarity, cosine distance, and reviewer overlap analysis.
- Implemented multiple filtering strategies to balance recommendation accuracy and diversity across domains.
- Outcome: Demonstrated improved cross-domain retrieval quality; formed the basis of CET research internship documentation.

Movie Genre Classifier (DistilBERT) — PyTorch, Hugging Face Transformers, NLP

- Fine-tuned DistilBERT on large-scale movie plot dataset for multi-label genre classification using Hugging Face Transformers and PyTorch.
- Implemented full pipeline: tokenization, training, evaluation, and inference; deployed a real-time web prediction interface.

Customer Churn Prediction System — Gradient Boosting, Scikit-Learn, Streamlit

- Built and deployed an interactive Streamlit dashboard for real-time churn prediction by analyzing demographics, account activity, and financial engagement patterns.
- Built an interactive Streamlit dashboard providing churn probability scores and actionable retention insights.

SMS Spam Detection System — *SVM, TF-IDF, NLP*

- Trained an SVM classifier on TF-IDF features for binary spam/ham classification; deployed a real-time web application for message analysis.

Content-Based Movie Recommendation Engine — *Python, Scikit-Learn, Cosine Similarity*

- Built a personalized recommendation engine using movie metadata (genres, descriptions, keywords) with similarity-based ranking.

EDUCATION

B.Tech – Computer Science & Engineering | LBS Institute of Technology for Women*Sep 2024 – 2028*

- CGPA: S1: 9.47 | S2: 9.43 | S3 : 8.7

Class 12 – Computer Science | Nirmala Bhavan Higher Secondary School*2022 – 2024*

- Score: 99.5% in Computer Science

TECHNICAL SKILLS

Languages	Python, C, C++, HTML, CSS
ML / DL	Scikit-Learn, PyTorch, Hugging Face Transformers, TensorFlow (basics)
NLP	TF-IDF, SVM Classifiers, Transformer fine-tuning (DistilBERT), Text Vectorization
Data & Tools	Pandas, NumPy, Streamlit, Git, GitHub, Data Visualization
Research	Technical Writing, Literature Review, Structured Experimentation, Embedding-Based Modeling

CERTIFICATIONS & ACTIVITIES

- NASA International Space Apps Challenge 2024 — "Galactic Problem Solver" certificate
- Google AI Essentials — Google (Coursera)
- Introduction to AI and ML on Google Cloud — Google Cloud (Coursera)
- AWS AI and ML Scholars 2026 Challenge — Amazon Web Services
- Data Science with Python Workshop — Techmaghi / IIT Hyderabad (Elan & nVision)
- Digital 101 (30 hrs) — NASSCOM IT-ITeS competency standard, Government approved, GOLD grade
- Google Solutions Challenge — Hack2Skill participant